

PROGRESS REVIEW

ESN-Based Battery SOC/SOH Estimation with Embedded Hardware Validation

Software Estimator + Embedded / FPGA Hardware Validation

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Problem & Proposed Solution

Why existing SOC/SOH estimation falls short, and what we propose instead

THE PROBLEM



Coulomb Counting

Open-loop integration accumulates drift and error over time.



Extended Kalman Filter (EKF)

Strongly model-dependent and computationally heavy for embedded use.

PROPOSED SOLUTION

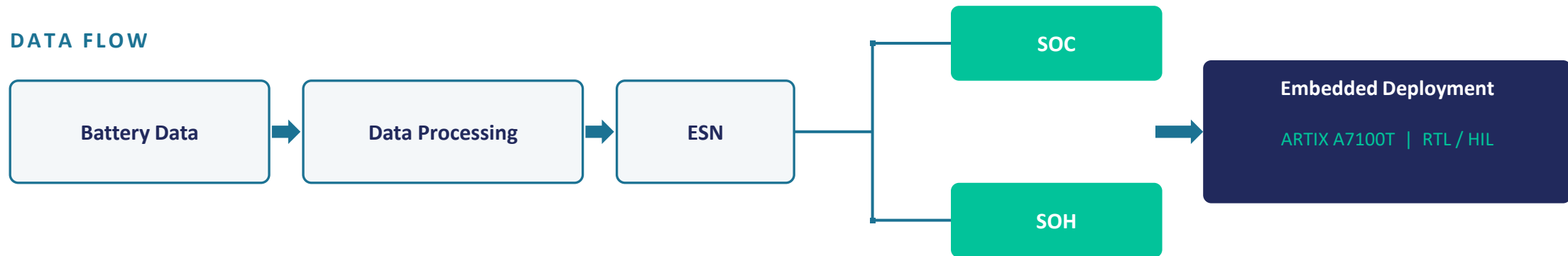
Echo State Network (ESN)

A lightweight reservoir-computing estimator for battery SOC/SOH — trained once, cheap to run, and well suited to embedded and hardware deployment.

Software = final estimator / product

Hardware = embedded implementation & verification platform

DATA FLOW



Phase 1 — Completed

Software estimator and embedded hardware, both verified

BATTERY / SOFTWARE

- ✓ 2-RC battery simulator
- ✓ EKF + Coulomb Counting + SOH tracking
- ✓ ESN training pipeline
- ✓ Flask dashboard deployed
- ✓ C99 ESN implementation
- ✓ CSR optimization — 6.7× speedup

EMBEDDED / HARDWARE

- ✓ Q6.10 fixed-point ESN datapath
- ✓ Hardware tanh LUT (odd-symmetry, positive half only)
- ✓ Verilog ESN RTL — 100-neuron reservoir
- ✓ BRAM-based weight / input / state storage
- ✓ Double-buffered recurrent state memory
- ✓ $Win \cdot u + W \cdot x$ MAC $\rightarrow$ bias $\rightarrow$ clip $\rightarrow$ tanh pipeline
- ✓ Vivado / XSim two-pass verification (ARTIX A7100T target)

200 / 200 neuron updates matched bit-exactly against the Python golden model

Phase 2 — In Progress

Current work on the ESN estimator and its hardware validation

ESN / SOFTWARE

- Online ESN adaptation using RLS / gradient descent
- LSTM / GRU benchmarking
- Testing across different EV drive cycles
- C99 deployment refinement

HARDWARE / EMBEDDED

- 1 Multi-timestep sequence input**
Extend the verified single-step design (reused $u(0)$) to full sequences $u(0) \dots u(T)$.
- 2 FPGA sequence-level verification**
Python golden model vs. Vivado/XSim, bit-exact comparison, on ARTIX A7100T.
- 3 ARTIX A7100T deployment**
On-board resource utilization and inference latency profiling.
- 4 Hardware-in-the-loop interface**
UART link between the battery simulator and the FPGA board.
- 5 End-to-end estimation**
Drive-cycle data $\rightarrow$ ESN $\rightarrow$ SOC / SOH on hardware.

Phase 3 — Final Goal & Timeline

Final goal: Deploy and validate a lightweight ESN-based SOC/SOH estimator on embedded hardware (ARTIX A7100T) using real / representative battery drive-cycle data.

FINAL ARCHITECTURE



TIMELINE

